

Homework 1

Name: _____

ID: _____

Q1 (20%) Multiple choice questions. Each question may have one or multiple answers.

- (1) For a convolution kernel with kernel size 3, stride 1, what is the zero padding number to keep the output feature map size unchanged?
 - (A) 0
 - (B) 1
 - (C) 2
 - (D) 3
- (2) Why AlexNet is large in size, but small in operations?
 - (A) Special FC layers
 - (B) Special Conv layers
 - (C) More channels
 - (D) Some redundant operators
- (3) In memory-efficient convolution (MEC), for an $n \times n$ kernels, at most how many times an input feature map value will be duplicated?
 - (A) n
 - (B) $2n$
 - (C) n^2
 - (D) n^3
- (4) The purpose/motivation of systolic array design is to:
 - (A) Make on-chip computation faster
 - (B) to load more data from memory
 - (C) Enable pipeline in matrix computation
 - (D) Reuse all loaded data
- (5) Which of the following is NOT included in Sparse convolutions?
 - (A) Processing non-zero elements
 - (B) Reducing memory footprint
 - (C) Utilizing dense connections between layers
 - (D) Decreasing computational requirements
- (6) Which of the following matrix decomposition methods can be used for accelerating DNNs?
 - (A) SVD
 - (B) CP Decomposition
 - (C) Tucker Decomposition
- (7) What is the advantage of strassen algorithm?

- (A) Better computation complexity
- (B) Better memory access pattern
- (C) Better computation overhead

Q2 (15%) In an embedding system, you need to implement a convolutional neural network inference. The system memory is limited to 2MB. For simplicity, assume the input feature map, kernel, lowered matrix and output feature map are all stored in the memory. Given the following convolution layer parameters.

Input Feature Map:

- batch size (N) = 4
- input channels (C) = 16
- feature map size (H × W) = 32 × 32

Convolution Kernel:

- output channels (K) = 32
- kernel size (F × F) = 3 × 3
- stride (S) = 1
- padding (P) = 1

Data Type: float32 (4 bytes/element)

Note: 1 MB = 1024 KB, 1 KB = 1024 bytes

- (a) For im2col method, calculate the size of the lowered matrix (rows × columns), the required extra memory size (in bytes). Is this implementation feasible? Why?
- (b) For memory-efficient convolution (MEC) method, calculate the size of the lowered matrix (rows × columns), the required extra memory size (in bytes). Is this implementation feasible? Why?
- (c) If the system supports 4 parallel threads, how would each method (im2col and MEC) utilize these threads in your opinion?

Q3 (15%) Given input feature map $\mathbf{X} = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$, convolution kernel $\mathbf{W} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$,

stride is 1 and no padding:

- (a) After applying img2col, the convolution is transformed into GEMM with the form of $\hat{\mathbf{X}} \times \hat{\mathbf{W}}$. Write down $\hat{\mathbf{X}}$ and $\hat{\mathbf{W}}$. What kind of sparse matrix representation should be used for $\hat{\mathbf{X}}$ and $\hat{\mathbf{W}}$, respectively? Write down the encoding result.
- (b) After applying img2col, the convolution is transformed into GEMM with the form of $\hat{\mathbf{W}} \times \hat{\mathbf{X}}$. Write down $\hat{\mathbf{X}}$ and $\hat{\mathbf{W}}$. What kind of sparse matrix representation should be used for $\hat{\mathbf{X}}$ and $\hat{\mathbf{W}}$, respectively? Write down the encoding result.
- (c) Which kind of img2col implementation is better? Please explain the reasons.

- Q4** (10%) The input feature map is shown in Figure 1, A has pixel value of 1 in 3 channels. For the convolution kernel settings, the kernel size is 3, number of output channel is 2, stride is 1 and no padding. Please implement the rulebook of the submanifold sparse convolution method to speed up the sparse computation.

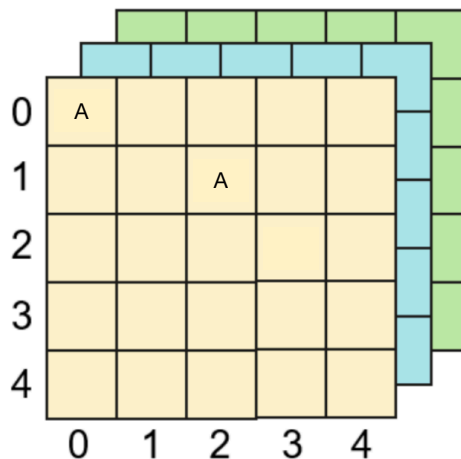


Figure 1: Sparse input feature map

- Q5** (15%) Given matrix $\mathbf{A} = \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix}$:
- Calculate the eigenvalues of the given matrix $\mathbf{A}$.
 - Please prove: $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{A} \mathbf{A}^\top$ are with the same eigenvalues.
 - Given the initial guess $\mathbf{x}_0 = [1 \ 1]$. Use the iterative method introduced in the class to calculate eigenvectors for matrix $\mathbf{A}$ (write down two iterative steps, if you can not provide the exact answer, at least describe the idea of the iterative method).
- Q6** (15%) In model pruning problem, Lasso regression is commonly used to compress the model with sparsity:

$$\mathcal{L}(\mathbf{w}) = \sum_{i=1}^n (y^i - \mathbf{w} \mathbf{x}^i)^2 + \lambda \|\mathbf{w}\|_1, \quad (1)$$

where $\mathbf{x}^i \in \mathbb{R}^d$, $y \in \mathbb{R}$ and $\mathbf{w} \in \mathbb{R}^d$.

- Calculate $\nabla \mathcal{L}(\mathbf{w})$.
- Can we use L-0 norm to add sparsity, such as $\mathcal{L}(\mathbf{w}) = \sum_{i=1}^n (y^i - \mathbf{w} \mathbf{x}^i)^2 + \lambda \|\mathbf{w}\|_0$? If yes, please give an example with brief illustration. If no, please illustrate the reason.
- Can we use gradient-based method to solve the the above quesiton with L-0 norm? If no, please provide rough ideas if other methods can be used to solve the L-0 norm optimization problem.
- Provide the closed-form solution to the single-variable Lasso problem:

$$\min_x \left\{ \frac{1}{2} (x - w)^2 + \lambda |x| \right\}. \quad (2)$$

Q7 (10%) Consider the **Lasso regression problem**:

$$\arg \min_{\boldsymbol{\beta}} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1. \quad (3)$$

- (a) **Group Lasso regularization.** Suppose that the coefficient $\boldsymbol{\beta}$ has **10** elements: $\beta_1, \dots, \beta_{10}$. If we choose $\beta_1, \dots, \beta_5$ to be Group_1, and $\beta_6, \dots, \beta_{10}$ to be Group_2. How do you formulate the group Lasso regression problem?
- (b) Explain how Group Lasso differs from standard Lasso regression and how it promotes structured sparsity. Provide a scenario in neural network pruning where Group Lasso regularization would be beneficial.